Torsional Rigidity of a Cubic Frame

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Introduction

Torsional rigidity describes a structure's resistance to twisting when an external torque is applied. A common way of increasing the rigidity of a frame is to add diagonal members, which divide rectangular faces into triangular sections and restrict deformation.

The purpose of this experiment was to compare the torsional behaviour of two similar cubic frames: one with diagonal trusses and one without them.

One face of each cube was fixed while the opposite face was loaded using rubber bands. A mirror and laser pointer were used to measure the resulting rotation.

The experiment also demonstrates an important structural design principle: significant increases in rigidity can be achieved by adding a small amount of material in strategically important locations. This allows lightweight structures to be made considerably stiffer without a large increase in weight.

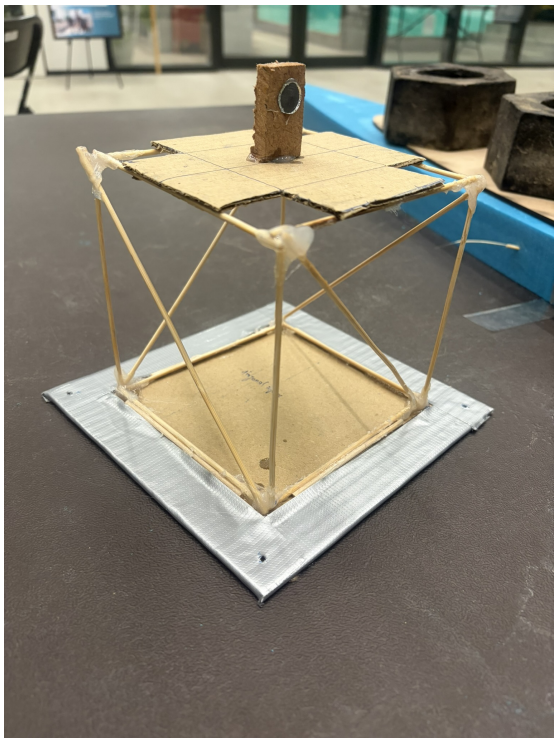


Figure 1: Completed test cube with diagonal trusses.

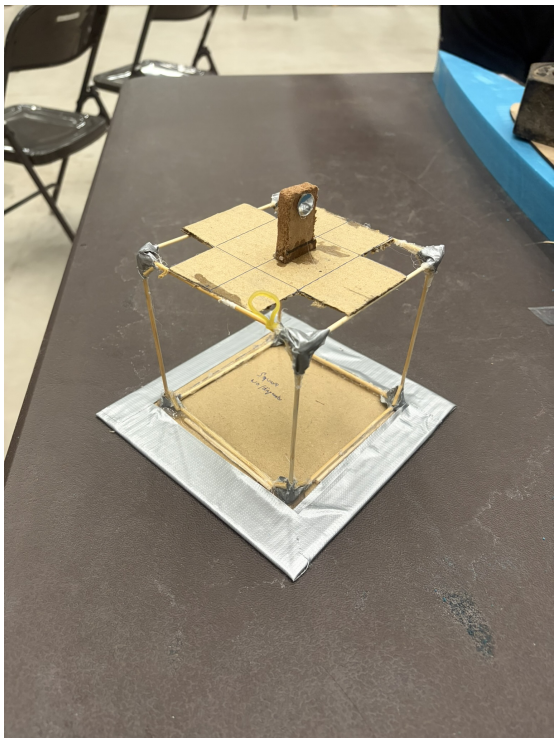


Figure 2: Completed test cube without diagonal trusses.

Method

Two small cubic structures were constructed for the experiment. Both structures were designed to have approximately the same dimensions and were made using similar lightweight materials.

The structures were made using wooden skewers and cardboard elements, with hot glue used to hold the members together.

The first structure was reinforced using diagonal trusses on the faces of the cube. The second structure was constructed without these diagonal trusses. The purpose of using these two configurations was to isolate the effect of diagonal bracing on the torsional deformation of the structure.

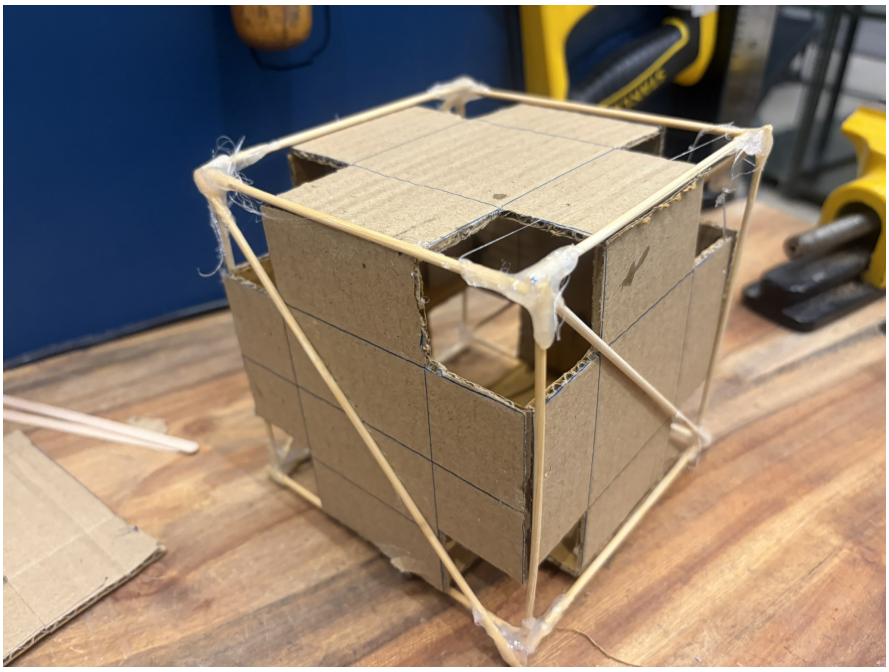


Figure 3: Construction of the cubic frame with diagonal trusses. Inner cardboard serve as a jig to help glue the vertices.

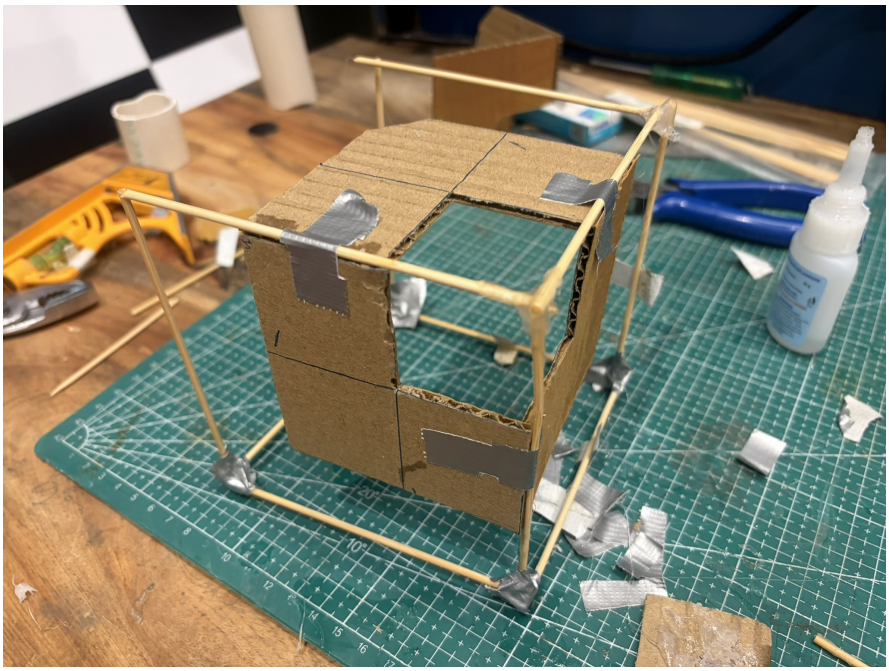


Figure 4: Construction of the cubic frame without diagonal trusses.

One face of each cube was rigidly fixed to a base. The opposite face was used as the loading face. The loading face was connected to two rubber bands which were positioned approximately tangentially to the structure. When the rubber bands were stretched, they

exerted forces on the loading face and produced a torque about the axis of the structure.

A small mirror was attached to the loading face. A laser pointer was positioned such that the laser beam was reflected from the mirror towards a distant wall.

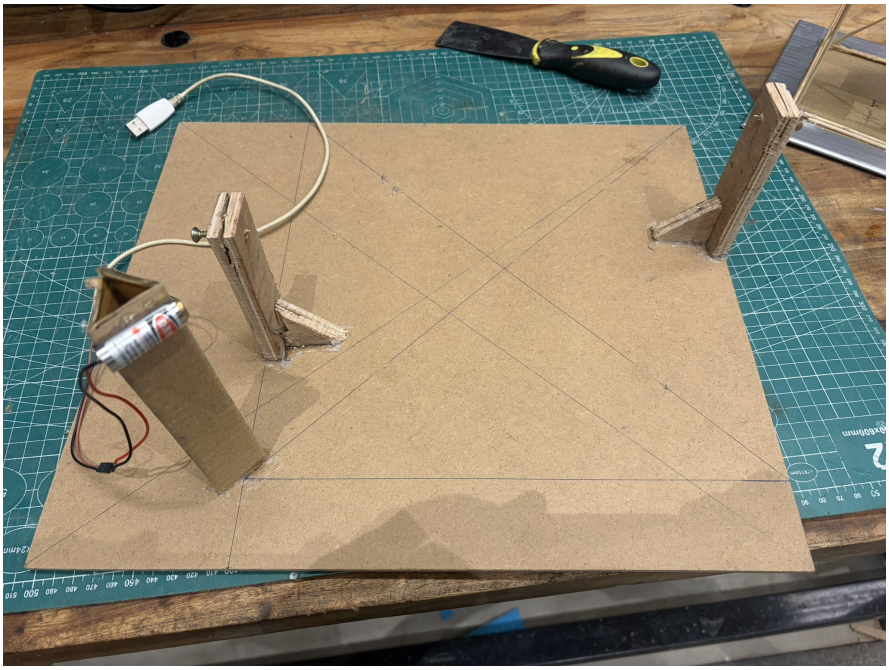


Figure 5: Experimental base

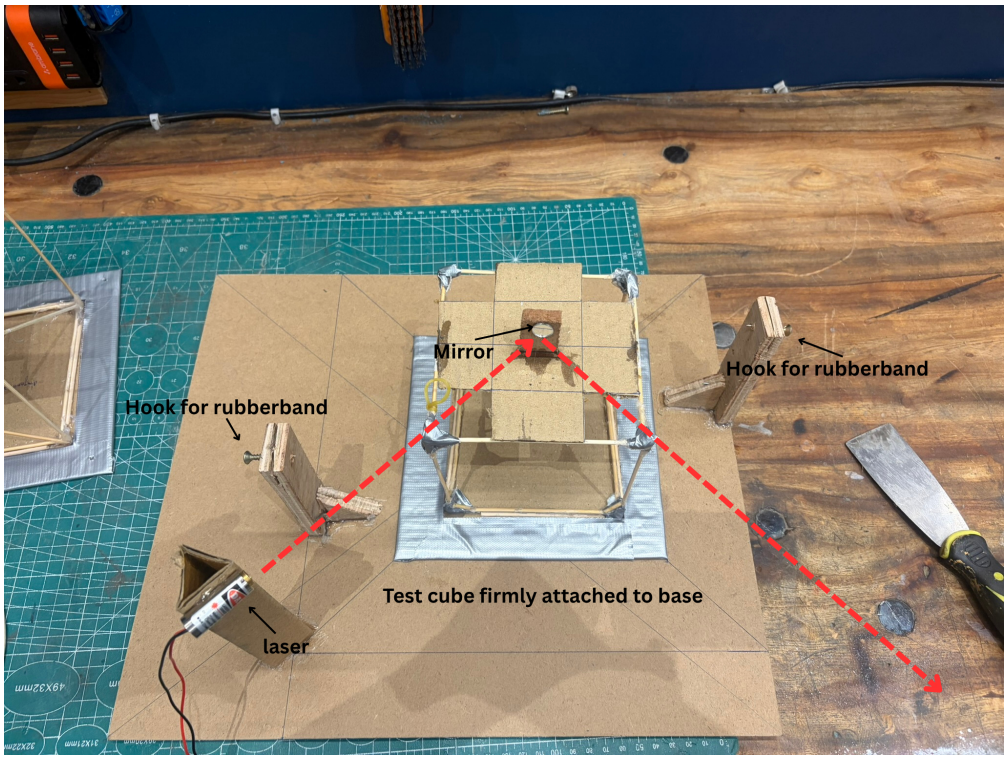


Figure 6: Overall experimental setup showing the test structure, supports and loading arrangement.

The distance travelled by the laser spot on the wall was used as an indirect measurement of the angular deformation of the loading face. Since the mirror rotates together with the loading face, rotation of the structure causes a change in the direction of the reflected laser beam.

The experiment was performed separately for the trussed and untrussed structures. Multiple measurements were taken for each configuration in order to reduce the influence of measurement variations.

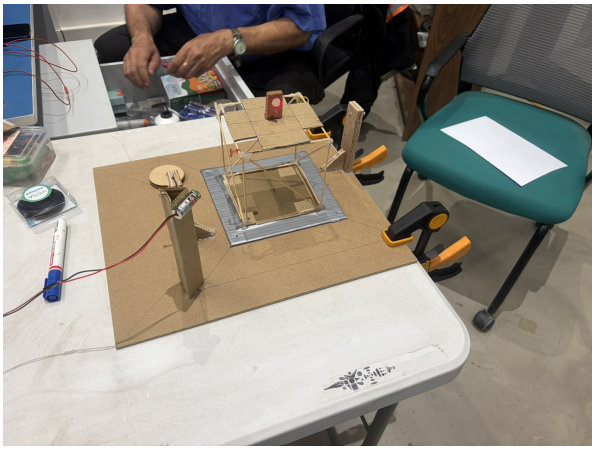


Figure 7: Experimental setup positioned on the working table.

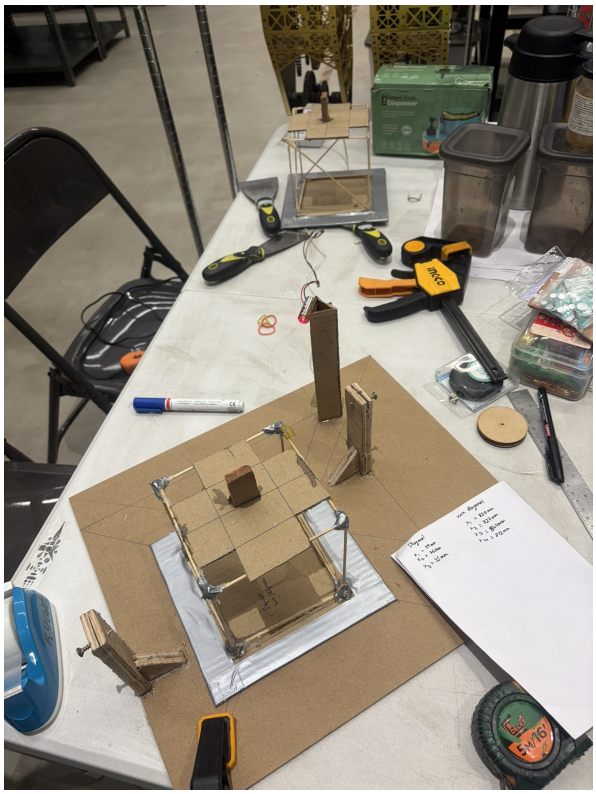


Figure 8: Alternative view of the experimental setup on the table.

Loading Method

The torsional loading was generated using two rubber bands. The rubber bands were attached to the loading portion of the structure and pulled tangentially in opposite directions.

The forces generated by the rubber bands produced a moment about the central axis of the cubic structure. This caused the loading face to rotate relative to the fixed face.

The exact force produced by the rubber bands was not measured directly during the experiment. Therefore, the main purpose of the experiment was to compare the relative deformation of the two structures rather than to determine an absolute torsional stiffness in SI units.

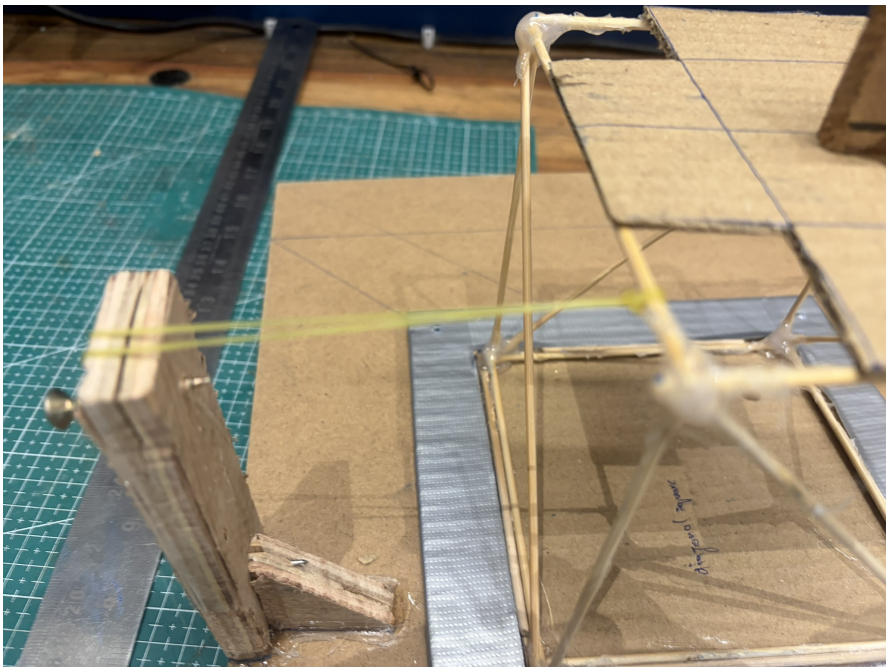


Figure 9: Attachment of the rubber bands to the loading structure.

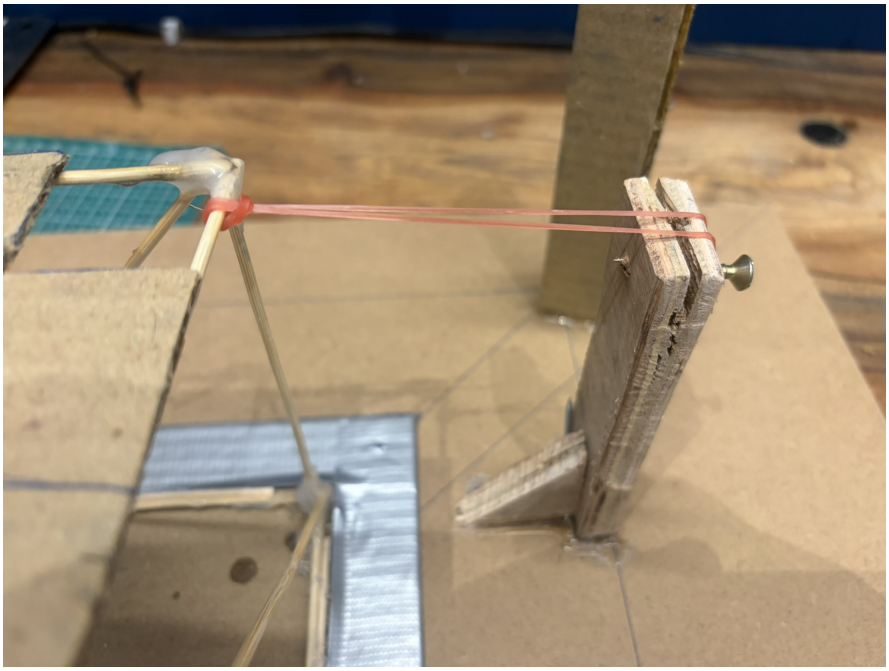


Figure 10: Close-up view of the rubber-band loading arrangement.

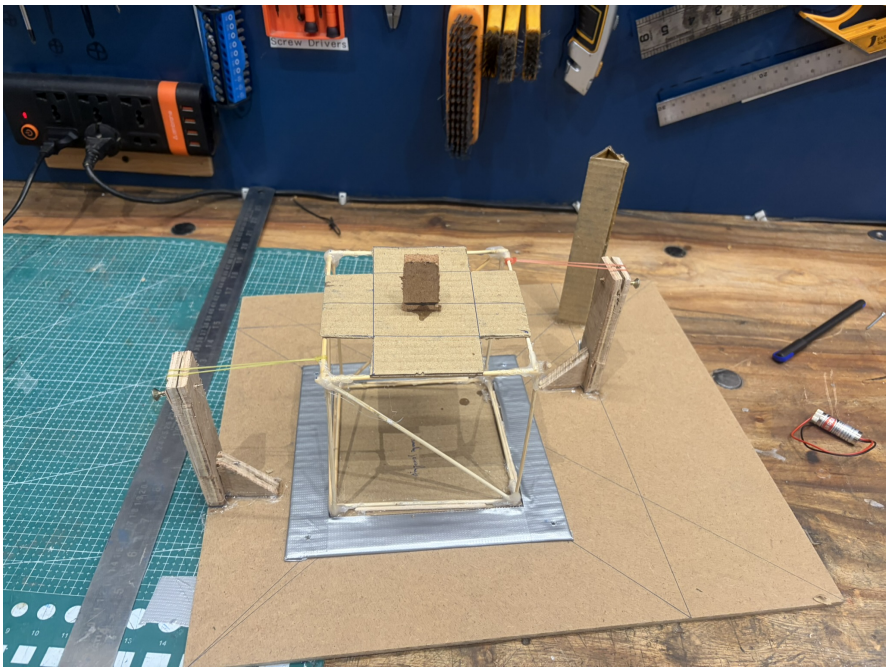


Figure 11: Torsional loading of the structure using rubber bands.

Laser Measurement Method

The laser and mirror arrangement was used to amplify the small angular rotation of the loading face.

When the mirror rotates through a small angle, the reflected laser beam changes direction by approximately twice the angle of rotation of the mirror. Since the beam travelled a relatively large distance before reaching the wall, even a small rotation of the structure resulted in a measurable displacement of the laser spot.

A reference position was first marked on the wall before applying the torsional load. The structure was then loaded and the new position of the laser spot was marked. The distance between the two positions was measured.

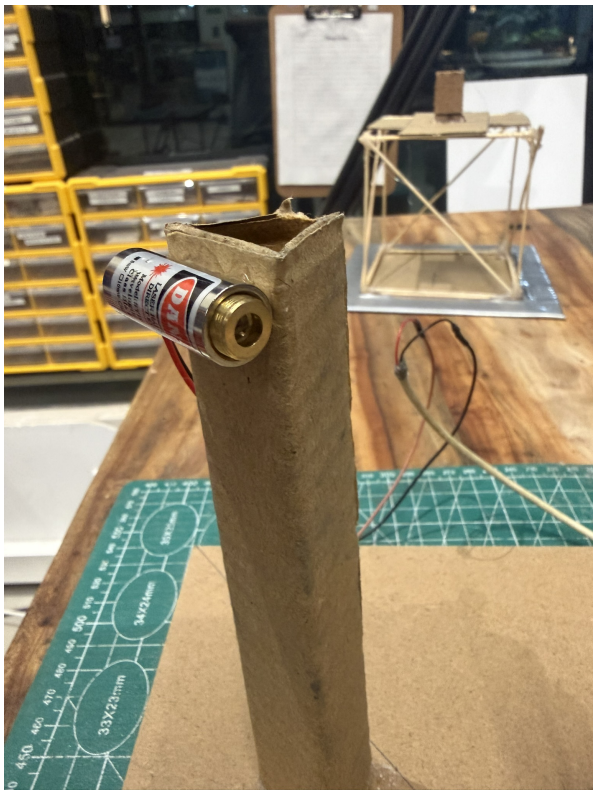


Figure 12: Laser setup on a pole



Figure 13: Laser spot measurements recorded on the distant wall.

The distance between the mirror and the wall was measured as approximately 9.75 m. The large distance between the experimental structure and the wall increased the measurable displacement of the reflected laser spot.

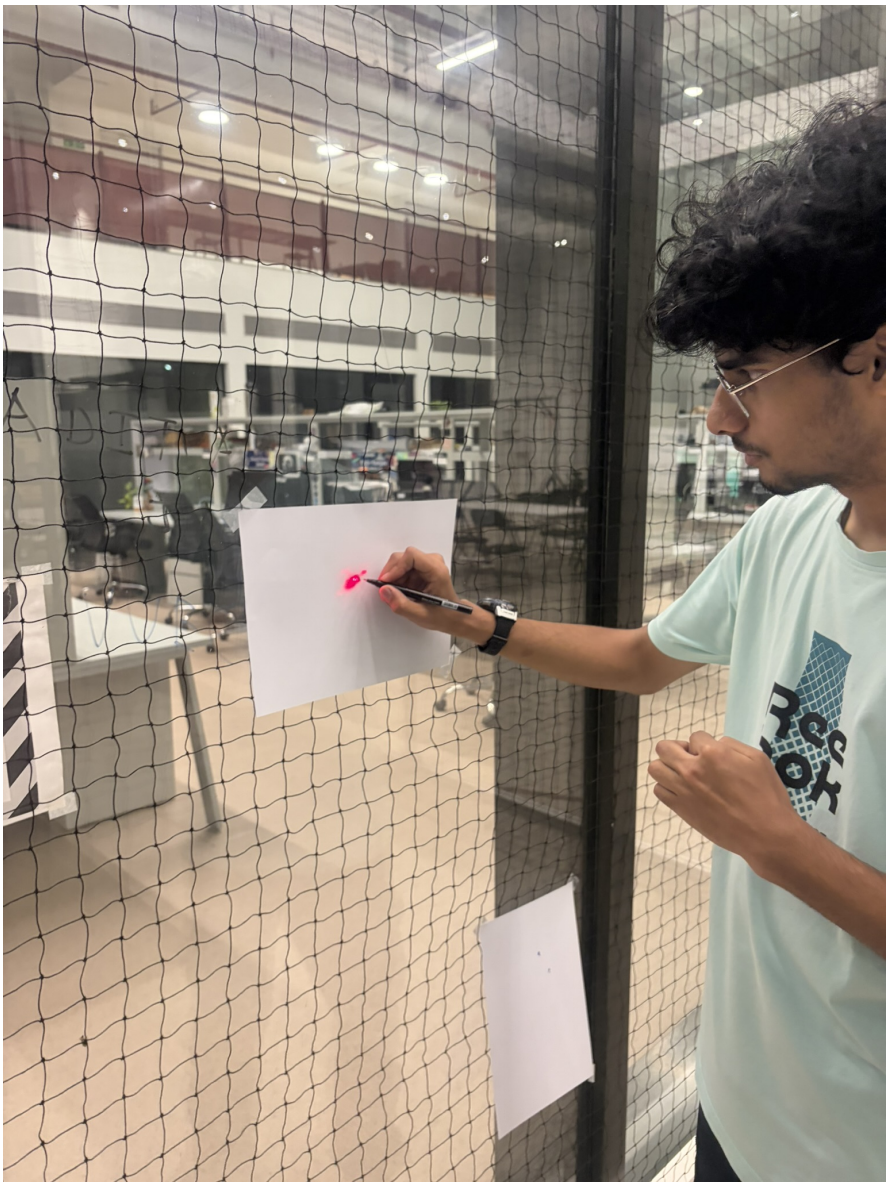


Figure 14: Marking the initial and displaced laser positions on the wall.

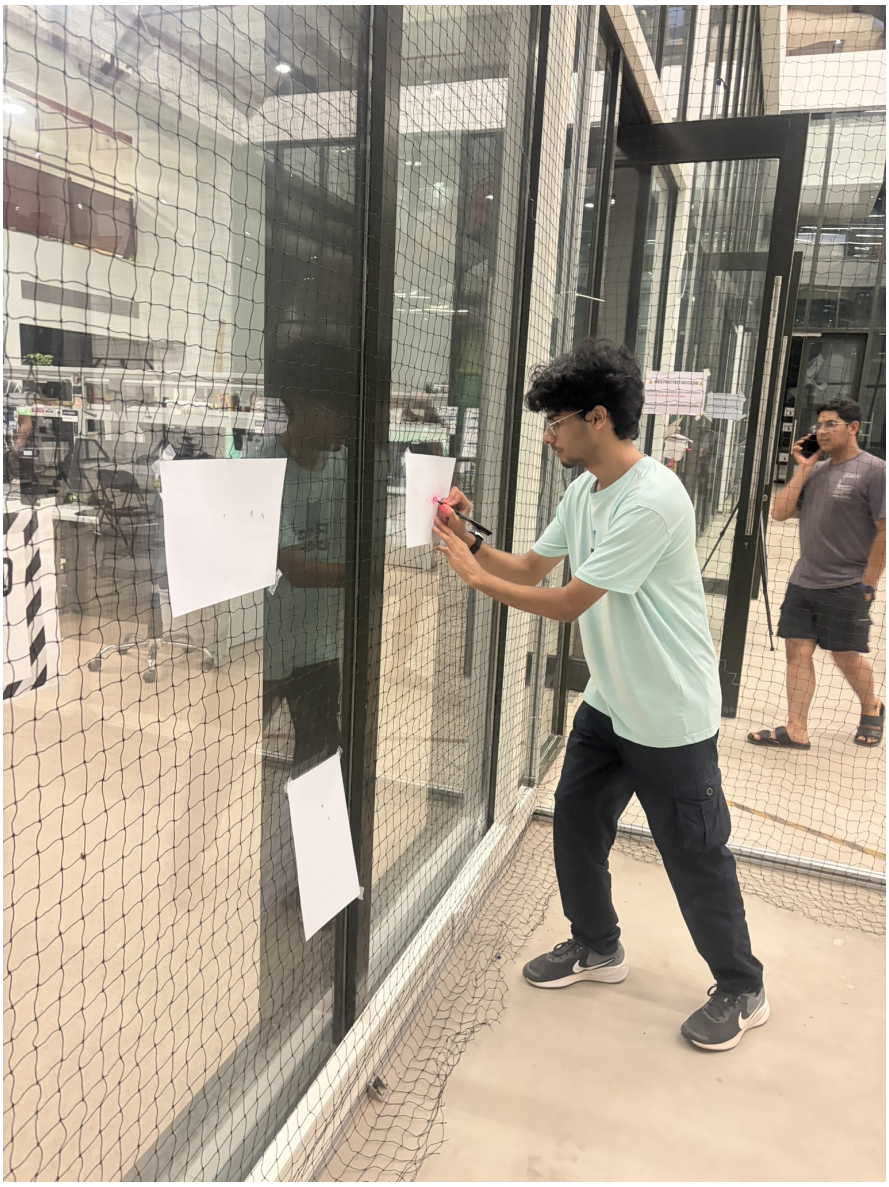


Figure 15: Measurement of the laser displacement on the wall.



Figure 16: Recording the laser displacement for an experimental trial.



Figure 17: Additional laser displacement measurement.



Figure 18: Final laser displacement measurement used during the experiment.

Experimental Procedure

The experiment was carried out using the following procedure:

1. The first cubic structure was placed on the experimental base.
2. The base was secured to the table using clamps.
3. One face of the structure was rigidly fixed to the base.
4. A mirror was attached to the loading face.
5. The laser pointer was aligned with the mirror so that the reflected beam reached the distant wall.

6. The initial position of the laser spot was marked on the wall.
7. The rubber bands were attached to the loading face.
8. The torsional load was applied using the rubber bands.
9. The new position of the laser spot was marked.
10. The distance between the initial and final laser positions was measured.
11. The experiment was repeated for three trials.
12. The same procedure was then carried out using the second cubic structure.
13. The measurements obtained from the two structures were compared.
14. The distance to the wall was measured and recorded for use in calculating the angular deformation of the loading face.

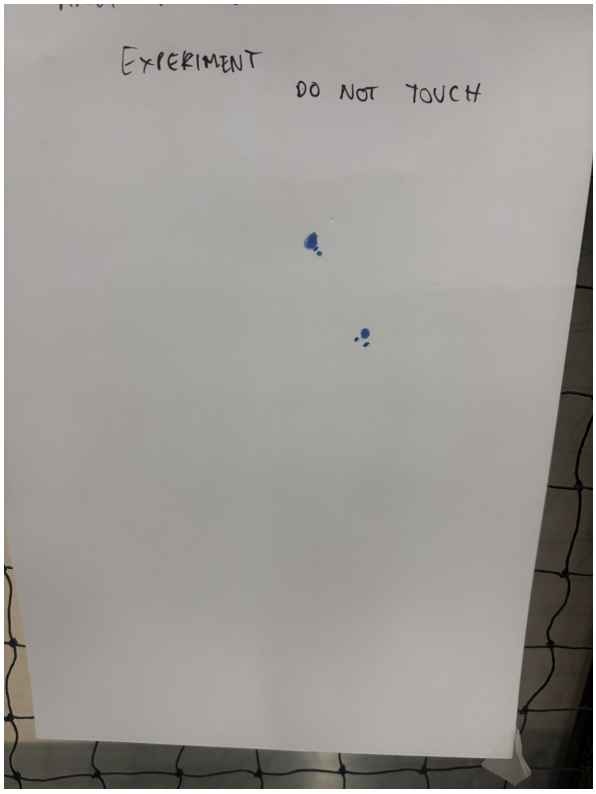


Figure 19: Laser displacement measurement for the trussed structure. It fit on a single A4 sheet. When its both loaded and unloaded.

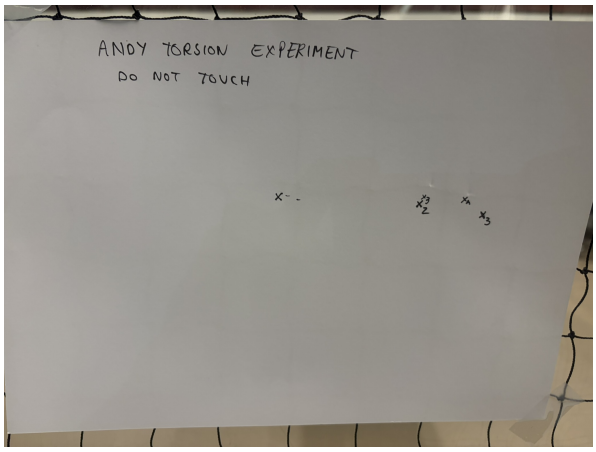


Figure 20: Starting laser point measured on an A4 sheet for un-trussed structure when unloaded.

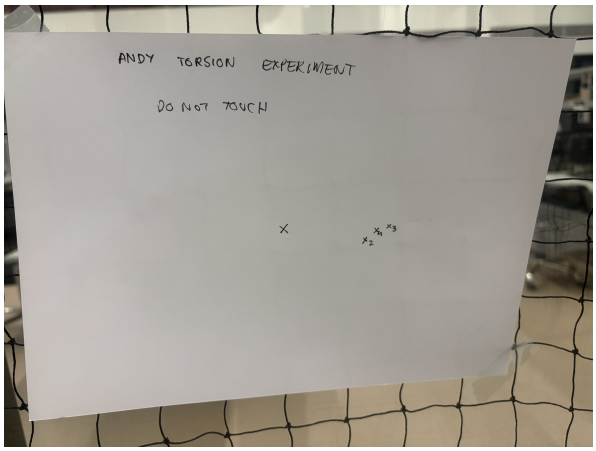


Figure 21: Final laser point measured on another A4 sheet for un-trussed structure when loaded.

Results

The measured displacement of the laser spot was recorded for both structural configurations.

The raw experimental measurements are shown in Table 1.

Table 1: Measured laser displacement for the two test structures.

Structure	Trial	Laser displacement (mm)
Trussed	1	37
Trussed	2	36
Trussed	3	35
Untrussed	1	850
Untrussed	2	827
Untrussed	3	840

The mean displacement for each structure was calculated from the three experimental measurements.

The following quantities were calculated from the experimental data:

- Mean laser displacement for the trussed structure.
- Mean laser displacement for the untrussed structure.
- Difference between the two mean displacements.
- Angular deformation of the loading face.
- Relative deformation between the two structures.
- Relative increase in effective torsional stiffness.

Calculations

The experimental calculations were performed using the measured laser displacements for the two structures. The distance between the mirror and the wall was measured as

$$L = 9.75\,\mathrm{m} = 9750\,\mathrm{mm}.$$

Mean Laser Displacement

For the structure with diagonal trusses, the measured laser displacements were

$$x_1 = 37\,\mathrm{mm}, \qquad x_2 = 36\,\mathrm{mm}, \qquad x_3 = 35\,\mathrm{mm}.$$

The mean displacement is therefore

$$\bar{x}_T = \frac{x_1 + x_2 + x_3}{3}$$

$$\bar{x}_T = \frac{37 + 36 + 35}{3}$$

$$\boxed{\bar{x}_T = 36.0\,\text{mm}}$$

For the structure without diagonal trusses, the three measured displacements used were

$$x_1 = 850\,\text{mm}, \qquad x_2 = 827\,\text{mm}, \qquad x_3 = 840\,\text{mm}.$$

The mean displacement is

$$\bar{x}_U = \frac{x_1 + x_2 + x_3}{3}$$

$$\bar{x}_U = \frac{850 + 827 + 840}{3}$$

$$\boxed{\bar{x}_U = 839.0\,\text{mm}}$$

Therefore, the mean laser displacement was

$$\boxed{\bar{x}_T = 36.0\,\text{mm}}$$

for the trussed structure and

$$\boxed{\bar{x}_U = 839.0\,\text{mm}}$$

for the untrussed structure.

Difference in Mean Displacement

The difference between the mean laser displacements of the two structures is

$$\Delta x_{\text{difference}} = \bar{x}_U - \bar{x}_T$$

$$\Delta x_{\text{difference}} = 839.0 - 36.0$$

$$\boxed{\Delta x_{\text{difference}} = 803.0\,\text{mm}}$$

Thus, the laser spot travelled an additional 803.0 mm for the structure without diagonal trusses.

Angular Deformation

The mirror was used as an optical lever to measure the angular rotation of the loading face. When the mirror rotates through an angle θ, the reflected laser beam rotates through approximately 2θ.

The relationship between the measured displacement of the laser spot and the angular rotation is therefore

$$\Delta x = L \tan(2\theta).$$

Hence,

$$\theta = \frac{1}{2}\tan^{-1}\left(\frac{\Delta x}{L}\right).$$

For the structure with diagonal trusses,

$$\theta_T = \frac{1}{2}\tan^{-1}\left(\frac{36.0}{9750}\right)$$

$$\boxed{\theta_T = 0.001846\,\text{rad}}$$

Converting this value to degrees,

$$\theta_T = 0.001846\left(\frac{180}{\pi}\right)$$

$$\boxed{\theta_T = 0.1058^\circ}$$

For the structure without diagonal trusses,

$$\theta_U = \frac{1}{2}\tan^{-1}\left(\frac{839.0}{9750}\right)$$

$$\boxed{\theta_U = 0.04292\,\text{rad}}$$

and therefore

$$\theta_U = 0.04292\left(\frac{180}{\pi}\right)$$

$$\boxed{\theta_U = 2.459^\circ}$$

The structure without diagonal trusses therefore experienced a considerably larger angular deformation than the structure with diagonal trusses.

Relative Deformation

The relative angular deformation of the two structures can be compared using

$$R_\theta = \frac{\theta_U}{\theta_T}.$$

Substituting the calculated values,

$$R_\theta = \frac{0.04292}{0.001846}$$

$$\boxed{R_\theta \approx 23.25}$$

Thus, the angular deformation of the structure without diagonal trusses was approximately 23.25 times greater than that of the structure with diagonal trusses.

Relative Increase in Torsional Stiffness

The effective torsional stiffness of a structure is defined as

$$k_t = \frac{T}{\theta}$$

where T is the applied torque and θ is the angular deformation.

Assuming that approximately the same torque was applied to both structures, the ratio of their torsional stiffnesses is

$$\frac{k_T}{k_U} = \frac{T/\theta_T}{T/\theta_U}.$$

The torque therefore cancels out, giving

$$\frac{k_T}{k_U} = \frac{\theta_U}{\theta_T}.$$

Using the calculated angular deformations,

$$\frac{k_T}{k_U} = \frac{0.04292}{0.001846}$$

$$\boxed{\frac{k_T}{k_U} \approx 23.25}$$

Therefore, under the assumption that the same torque was applied to both structures, the structure containing diagonal trusses had an effective torsional stiffness approximately

$$\boxed{23.25 \text{ times}}$$

that of the structure without diagonal trusses.

The corresponding percentage increase in effective torsional stiffness is

$$(23.25 - 1) \times 100$$

$$\boxed{\approx 2225\%}$$

relative to the untrussed structure.

It should be noted that the torque was generated using rubber bands and was not directly measured. Therefore, the calculated value represents a relative stiffness comparison based on the assumption that approximately the same torque was applied to both structures, rather than an absolute measurement of torsional stiffness.

Discussion

The experimental results show a clear difference between the torsional behaviour of the two cubic structures.

The structure without diagonal trusses experienced substantially greater deformation when subjected to the torsional loading. This indicates that the rectangular faces of the unbraced structure were able to distort relatively easily when the loading face was rotated.

In contrast, the structure containing diagonal trusses showed considerably less deformation. The diagonal members constrained the relative movement of the corners of the frame and created additional load paths through the structure.

The main structural mechanism responsible for this difference is triangulation. A rectangular frame can change its shape relatively easily under shear loading, whereas a triangular arrangement is geometrically constrained. By adding diagonal members, the faces of the cube become more resistant to distortion.

The laser measurement method also proved useful for the experiment. The actual angular rotation of the structure was relatively small, but the optical arrangement converted this small rotation into a much larger displacement of the laser spot on the wall. This made it possible to measure the deformation using a simple experimental setup.

Effect of Diagonal Trusses

The experimental comparison demonstrates that diagonal bracing can significantly improve the torsional rigidity of a frame structure.

The trusses provide additional structural members that carry loads primarily through axial tension and compression. This reduces the amount of deformation that must be accommodated through bending and distortion of the frame.

The results therefore support the expected relationship between structural triangulation and rigidity.

Sources of Experimental Error

There are several factors that could have affected the experimental measurements.

The force generated by the rubber bands was not directly measured. Since the force of an elastic band depends on its extension, small differences in the loading arrangement may have resulted in different applied torques.

The two structures were also manually fabricated, meaning that their dimensions, member stiffnesses, and connection quality may not have been perfectly identical.

The adhesive joints used in the structures may have introduced additional flexibility. The measured deformation therefore represents the behaviour of the complete model rather than only the theoretical deformation of the structural members.

There may also have been some uncertainty in determining the exact position of the laser spot on the wall. The size of the laser spot and the accuracy with which the marks were measured can both contribute to experimental uncertainty.

Finally, the observed deformation may not have consisted entirely of pure torsion. Bending, shear deformation, joint movement, and local deformation may also have contributed to the measured laser displacement.

Conclusion

The experiment investigated the effect of diagonal trusses on the torsional rigidity of a cubic frame.

Two similar structures were tested: one with diagonal trusses and one without diagonal trusses. One face of each structure was fixed while the opposite face was subjected to torsional loading using rubber bands.

A mirror and laser pointer were used to measure the resulting deformation. The movement of the reflected laser spot provided an amplified measurement of the rotation of the loading face.

The results showed that the structure without diagonal trusses underwent significantly greater deformation than the structure with diagonal trusses.

This demonstrates the important role of triangulation in increasing the rigidity of a frame. The diagonal members restrict the distortion of the rectangular faces and provide additional load paths through the structure.

Therefore, the experiment provides experimental evidence that adding diagonal trusses can substantially improve the resistance of a cubic frame to torsional deformation.

Future Work

The experiment could be improved by directly measuring the force applied by the rubber bands. A load cell or force sensor could be incorporated into the loading system so that the applied torque can be determined more accurately.

The dimensions and material properties of the two structures could also be controlled more precisely. This would reduce the differences between the specimens and make the comparison more reliable.

A larger number of loading levels could also be tested. Instead of applying only one loading condition, several different torque values could be applied and the corresponding angular deformation could be measured.

This would allow a torque versus angular deformation graph to be produced. The slope of this relationship could then be used to determine the effective torsional stiffness of each structure.

Different truss configurations could also be tested to determine how the number and arrangement of diagonal members influence the overall torsional rigidity of the structure.